

XGBoost Math.

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Regression

$$SS = \frac{(\sum \text{Residuals})^2}{N + \lambda}$$

$$\text{Output} = \frac{\sum \text{Residuals}}{N + \lambda}$$

Classification

$$SS = \frac{(\sum \text{Residuals})^2}{\sum [P(1-P)] + \lambda}$$

$$\text{Output} = \frac{\sum \text{Residuals}}{\sum [P(1-P)] + \lambda}$$

XGBoost Loss F^n :

$$\hat{y}_i = m_1 + m_2 + \dots$$

base decision tree.

$$f_1(x_i) + f_2(x_i) + \dots$$

$$\hat{y}_i = \sum_{i=1}^n f_i(x_i)$$

$$L = \underbrace{\sum_{i=1}^n L(y_i, \hat{y}_i)}_{\text{loss } f^n} + \underbrace{\Omega(f_2(x_i))}_{\text{regularization term}}$$

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \cdot \|w\|^2$$

regularization parameter (3)
leaf nodes.

Stage I: $\xrightarrow{\text{mean}}$

$$\boxed{m_1} \xrightarrow{\text{mean}} L^{(1)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i)$$

$$f_1(x_i)$$

Stage II:

$$\boxed{m_1} \quad \boxed{m_2} \xrightarrow{\text{mean}} L^{(2)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \Omega(f_2(x_i))$$

$$f_1(x_i) \quad f_2(x_i)$$

$$\hat{y}_i = f_1(x_i) + f_2(x_i)$$

Stage III:

$$\boxed{m_1} \quad \boxed{m_2} \quad \boxed{m_3} \xrightarrow{\text{mean}} L^{(3)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \Omega(f_3(x_i))$$

$$f_1(x_i) + f_2(x_i) + f_3(x_i)$$

Stage t:

$$\boxed{m_1} \quad \boxed{m_2} \quad \dots \quad \boxed{m_t} \xrightarrow{\text{mean}} L^{(t)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \Omega(f_t(x_i))$$

$$f_1(x_i) + f_2(x_i) + \dots + f_t(x_i)$$

$$L^{(t)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t(x_i))$$